

Artificial Intelligence...Sooner or Later – Beyond autonomous driving: What added value can SDVs deliver?



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This report is part of a global series in which our analysts consider what their sector will look like by the early 2030s when AI has been fully integrated and commercialized. There are broadly two schools of thought running through our research on AI globally. The “Sooner” camp generally believes that Artificial Intelligence is a game-changer for their sector, right now. The “Later” camp believes that Artificial Intelligence is a game-changer for their industry... just not yet. In this series, each analyst is declaring their view on the timing and nature of AI disruption in their sector and what that implies for warranted valuation today.

AI is replacing electrification as the primary differentiator: As EV momentum stalls, automakers are pivoting toward SDVs and intelligent mobility to sustain competitiveness. IVI and AD, core elements of these technologies, are driven by AI and the automotive sector broadly stands as a beneficiary. The deployment of E2E stacks and VLA models is accelerating the roadmap toward L4/L5 autonomy (~4% penetration by 2035), shifting the vehicle's value proposition from hardware performance to a platform for software intelligence and new added value. Post autonomy, the emergence of non-driving time creates new avenues for value creation. Now we identify three strategic directions: mobility as a portable home, mobility as a creative entertainment space, and robotaxi monetization.

3 strategies for enhancing value in non-driving time: Sharp and Xiaomi focus on expanding the vehicle's role as a portable home, integrating the cabin with the user's broader device ecosystem. Sharp's LDK+ emphasizes stationary utility and appliance connectivity, while Xiaomi's 'Human x Car x Home' vision uses HyperOS to unify the car with household devices. Sony Honda Mobility pursued a different path by framing mobility as an entertainment space. Although development of the Afeela 1 has been discontinued, we believe the concept could be revisited given Sony's strengths in content and audio visual technology. Tesla represents a third approach, aiming to leverage full autonomy to convert idle private vehicles into part of a shared robotaxi network.

Interior as the foundation for post autonomy value creation: As L4/L5 autonomy could enhance value in non-driving time, Toyota is starting to position interior space as a core differentiator. Concepts such as the e-Palette and the Lexus LS Concept highlight Toyota's focus on flexible layouts and quiet, comfortable cabins, supported by group suppliers like Toyota Boshoku and Toyoda Gosei moving toward integrated interior systems. We believe further consolidation among key suppliers may be needed to accelerate cockpit integration. However, Toyota lacks applications that utilize this interior environment, and building external partnerships to develop in-cabin services will become increasingly important.

Scale and data depth support future application expansion: Autonomy could shift value toward upstream AI architecture and downstream applications, gradually reducing differentiation in traditional manufacturing. Toyota's 150-mn-unit global fleet and the Mobility Services Platform provide access to large, diverse real-world data, which strengthens not only its AI development but also its potential for future external collaboration to expand in-cabin applications. Mid-sized players, by contrast, may face increasing difficulty in capturing downstream value and risk remaining hardware suppliers.

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	15 Apr 2026		TTM Rel. Perf.	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target		Cur	2024A	2025E	2026E	2024A	2025E	2026E
7203.JP (Toyota)	O	JPY	3,381.00	4,250.00	(15.1)%	JPY	359.56	302.42	346.01	9.4	11.2	9.8
7269.JP (Suzuki)	O	JPY	1,877.50	3,150.00	(36.3)%	JPY	215.65	213.28	230.73	8.7	8.8	8.1
7267.JP (Honda)	M	JPY	1,285.00	1,400.00	(56.4)%	JPY	178.93	107.82	163.70	7.2	11.9	7.8
7270.JP (Subaru)	U	JPY	2,506.00	2,800.00	(47.7)%	JPY	458.00	184.74	291.46	5.5	13.6	8.6
7201.JP (Nissan)	U	JPY	362.10	300.00	(36.6)%	JPY	(187.08)	(144.11)	46.67	(1.9)	(2.5)	7.8
7261.JP (Mazda)	U	JPY	1,078.50	1,000.00	(17.6)%	JPY	180.87	13.31	131.26	6.0	81.0	8.2
JPL			2,422.38									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

JAPAN AUTOS

We expect the roll out of SDVs by Japanese automakers to kick off with Toyota's new RAV4, set to launch in 2025. Japanese automakers are progressing on SDV development with varying strategies, ranging from full-stack software platform/OS development to focused development aligned with company strengths.

In autonomous driving, rapid advances in AI are accelerating the shift from traditional rule-based stacks to E2E stacks, driven by the need for faster deployment and improved handling of edge cases. Within E2E architectures, Vision-Language-Action (VLA) models, leveraging generative AI and LLM-based reasoning, are increasingly viewed as the next frontier. Toyota is pursuing a 'multi-pathway' strategy that combines in-house development through its smart-city initiative, Woven City, with external partnerships, including collaboration with Waymo, a pioneer in VLA-based autonomous driving, as well as strategic investments in May Mobility, Pony.ai, and Momenta. Crucially, its 150m-unit global fleet and Mobility Service Platform (MSPF) provide a structural advantage in aggregating the vast driving data required for model training. Honda, while working with Helm.ai, continues to focus primarily on developing its own in-house E2E autonomous driving system. Nissan, meanwhile, is leveraging Wayve's E2E technology as the core of its autonomous driving approach.

Finally, we expect only a limited number of automakers are likely to be positioned to deliver substantial added value once full autonomous driving is realized. Those with large global vehicle installed bases, scale advantages and sufficient investment capacity, such as Toyota, are the most likely candidates. On the other hand, mid-sized and smaller manufacturers may find it increasingly difficult to secure downstream value and maintain long-term competitiveness, as weakening value capture leaves them at risk of remaining just vehicle manufacturers.

With Japan's government positioning SDVs as a national priority and setting a 30% penetration target by 2030-2035, we expect strategic initiatives across the sector to accelerate, although the pace and direction of progress vary by automaker. We rate **Suzuki** and **Toyota** as **Outperform**, **Honda** as **Market-Perform**, **Nissan**, **Mazda** and **Subaru** as **Underperform**.

As the industry moves toward full AI-driven autonomous driving, we assess the extent to which our coverage Japanese automakers could benefit from AI across five dimensions. Specifically, we evaluate each company based on its R&D capacity to sustain AI/AD-related development, the current maturity of AD technologies, organizational readiness for software development, the ability to monetize applications and services, and the degree of supply chain integration (Exhibit 1).

Based on these criteria, we continue to view Toyota as the leading beneficiary of AI. We believe Toyota has the capacity to invest more than JPY 8 tn in R&D over the next five years, supported by its substantial cash on hand and strong cash generation. This financial strength provides sufficient headroom for in-house development of autonomous driving technologies. In addition to internal development, Toyota is pursuing a multi-pathway approach, spanning rule-based to E2E AD, through partnerships and equity investments. While concrete examples of post full-autonomy applications and monetization remain somewhat limited, Toyota's unparalleled installed base of ~150 mn vehicles on the road, combined with its MSPF, provides a strong foundation to roll out a wide range of services. Moreover, Toyota benefits from a robust group supply chain, including Denso for ADAS and AD development, as well as Toyoda Gosei and Toyota Boshoku in vehicle interiors, an area that is expected to become increasingly

important in the autonomous driving era.

We see Nissan as the second most likely beneficiary. Amid weak financial performance and a negative FCF margin, Nissan's R&D capacity is constrained at present. That said, through its partnership with Wayve, Nissan plans to deploy cutting-edge E2E AD technology in FY3/28. The company has also been actively pursuing robotaxi initiatives, including the EasyRide robotaxi service being piloted near its HQs, as well as partnerships with Uber and Wayve for robotaxi pilot service in Tokyo. Once the ongoing restructuring is completed, the key focus will be whether Nissan can demonstrate further progress on the application and monetization front.

Honda has a certain degree of investment capacity supported by its motorcycle business as a cash cow. However, its automotive business continues to struggle, and at this stage, we do not believe Honda is well positioned to benefit meaningfully from AI. In autonomous driving, Honda had planned to launch GM Cruise in early 2026, but this was canceled following the closure of Cruise. More recently, Honda announced the termination of development for Afeela, the flagship project of Sony Honda Mobility, which had been the only Japanese automaker actively pursuing VLA development. Progress in in-house E2E AD development seemingly lagged as well. Going forward, Honda will need to strengthen integration with Astemo, which was recently consolidated, and provide clearer updates on the status of internal development.

For Suzuki, Mazda, and Subaru, scale limitations relative to Toyota, Nissan, and Honda constrain their ability to pursue autonomous driving development independently. As autonomous driving is expected to become an increasingly important factor in vehicle purchasing decisions, these three manufacturers will need to articulate a clear strategic direction at an early stage. Key questions include whether to leverage external technologies, as Nissan has done, or to more fully utilize alliances with Toyota.

EXHIBIT 1: Overall assessment of AI benefit potential across our coverage Japanese automakers; Toyota as the leading beneficiary

	Toyota	Nissan	Honda	Suzuki	Subaru	Mazda
Overall assessment of AI benefit potential						
R&D Capacity						
R&D plan (FY3/27E-3/31E)	JPY 8.3 tn	JPY 3.2 tn	JPY 6.6 tn	JPY 1.7 tn	JPY 1.0 tn	JPY 0.9 tn
Auto net cash (as of Dec-25)	JPY 14 tn ^{*1}	JPY 958 bn	JPY 3.1 tn ^{*2}	JPY 156 bn ^{*3}	JPY 444 bn ^{*3}	JPY 188 bn ^{*3}
Auto FCF margin (FY3/25)	4.4%	-2.1%	4.6% ^{*2}	2.6%	1.9%	2.1%
AD tech maturity	'Multi-pathway' approach	Wayve's E2E AD stack	In-house development in collaboration with Helm.ai	No specific plan	No specific plan; developing proprietary L2 ADAS 'EyeSight'	No specific plan
Organizational readiness for S/W development	Established a S/W subsidiary 'Woven by Toyota' in 2018	Established multiple S/W-specialized hubs	Consolidate Astemo; Established multiple S/W-specialized hubs	Established a S/W-specialized hub in India with Tata Elxsi	Established a S/W-specialized hub 'SUBARU Lab'	Established a S/W-specialized hub 'Mazda R&D Center Tokyo'
Application/service monetization	Global collaborations such as Pony.ai robotaxi projects and a partnership with Waymo	Automaker-led monetization trials (Uber Collab, EasyRide)	Multiple trials, yet limited scalability with some initiatives discontinued (Cruise in Japan, Afeela)	Minimal focus	Minimal focus	Minimal focus
Supply chain integration	Robust supply chain with affiliated suppliers such as Denso	Certain degree of relationship with mega suppliers and Wayve	Robust supply chain with affiliated suppliers such as Astemo	Limited supply chain integration	Limited supply chain integration	Limited supply chain integration

Note: ^{*1} 'Net Liquid Assets' as of FY3/25; ^{*2} Including 2W business; ^{*3} Consolidated basis

Source: Company reports, Bernstein analysis and estimates

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AUTONOMOUS DRIVING AS THE FOUNDATION FOR NEW VALUE CREATION

As the momentum for EV adoption decelerates, the automotive industry is shifting its focus toward the added value provided by AI-driven Software Defined Vehicles (SDVs) and intelligent mobility. IVI (in-vehicle infotainment) and autonomous driving, which are core elements of these technologies, are also driven by AI and have been evolving rapidly in recent years. 2025 has emerged as a year of significant headwinds for EVs. In the US, the Trump administration has moved to cut EV subsidies, while in Europe, the previous commitment to a total ban on internal combustion engine vehicles by 2035 has been walked back. Even in China, the full exemption of the vehicle purchase tax ended, replaced by a 50 percent reduction. This shift suggests that electrification alone is no longer a sufficient differentiator, forcing automakers to pivot toward software intelligence to sustain their competitiveness.

While various sectors, such as SaaS, face growing concerns over AI-driven disruption, the automotive industry stands out as a primary beneficiary of the technology. The most significant impact of AI is seen in the autonomous driving domain, specifically through the deployment of End-to-End (E2E) stacks and Vision-Language-Action (VLA) models. These technologies are steadily advancing the roadmap toward the realization of L4 & L5 autonomy, effectively enhancing the vehicle's overall value proposition.

Levels of Autonomous Driving

Autonomous driving (AD) capabilities are categorized according to the SAE (Society of Automotive Engineers) classification system from Level 0 to Level 5 (Exhibit 2). Currently, Level 2 ADAS (Advanced Driver Assistance Systems) has become standard on many mass-production vehicles. Tesla (not covered) and several Chinese automakers (e.g., Huawei-backed HIMA, XPeng, Li Auto, BYD, etc.) are already offering highway NOA (Navigate on Autopilot) equivalent to Level 2+ and/or urban NOA capabilities in select areas, approximating Level 2+++. Furthermore, in December 2025, the Chinese government granted the first two L3 permits to BAIC Arxfox (Alpha S) and Changan Deepal (SL03), allowing them to operate on designated expressways in Beijing (up to 80 km/h) and Chongqing (up to 50 km/h) respectively.

Up to Level 2 autonomy, responsibility for the driving task remains with the human driver, and such systems are therefore classified as ADAS that support the driver, rather than replace it. From Level 3 onward, responsibility shifts to the system itself, marking entry into what is generally regarded as the domain of autonomous driving.

The classification of autonomous driving levels can be closely linked to the scope of the operational design domain (ODD). Level 3 autonomous driving allows eyes-off operation within a defined ODD, while still requiring that a human driver be available to intervene when requested. Level 4 extends this further, enabling autonomous operation within the ODD without any human involvement. At Level 5, the ODD constraint is removed entirely, allowing autonomous driving under all conditions without human intervention.

EXHIBIT 2: Levels of autonomous driving; globally, highway & urban NOA are becoming increasingly widespread

	L0	L1	L2	L2+ (Highway NOA)	L2++ (Urban NOA)	L3	L4	L5
Definition	Manual driving	Assisted driving	Partial automated driving	Partial automated driving in highway	Partial automated driving in urban	Partial automated driving	Fully autonomous with limited ODD	Fully autonomous with all ODDs
Responsibility	Driver					Car		
	ADAS (Advanced driver assistance systems)					AD (Autonomous driving)		
ODD	-	High	Mid-hi	Mid		Low	Little	All
	None	Limited						Anywhere
Feet-off	⊗	✔	✔	✔	✔	✔	✔	✔
Hands-off	⊗	⊗	✔	✔	✔	✔	✔	✔
Eyes-off	⊗	⊗	⊗	⊗	⊗	✔	✔	✔
Mind-off	⊗	⊗	⊗	⊗	⊗	⊗	✔	✔
Driver-off	⊗	⊗	⊗	⊗	⊗	⊗	⊗	✔
Driver Involvement	Always required	Continuous monitoring & hands on wheel required		Continuous monitoring & readiness to intervene required		Required only if system fails	Not required	
Description	• Driver performs all driving tasks • Only basic alerts or emergency warnings	• System controls speed or steering • Driver handles all other actions	• System managing speed and steering	• System handling lane change, merging, exits on highway • Driver intervenes when requested	• System handling lane change, merging, exits on surface streets • Driver intervenes when requested	• System drives in defined ODDs • Driver intervenes when requested	• System drives in geo-fenced areas • Driver does not need to respond • Limited ODD	• System drives in all conditions • No driver involvement required • Any road, any condition

Source: SAE, Dynamic Map Platform, Bernstein China Semiconductors team, Bernstein analysis

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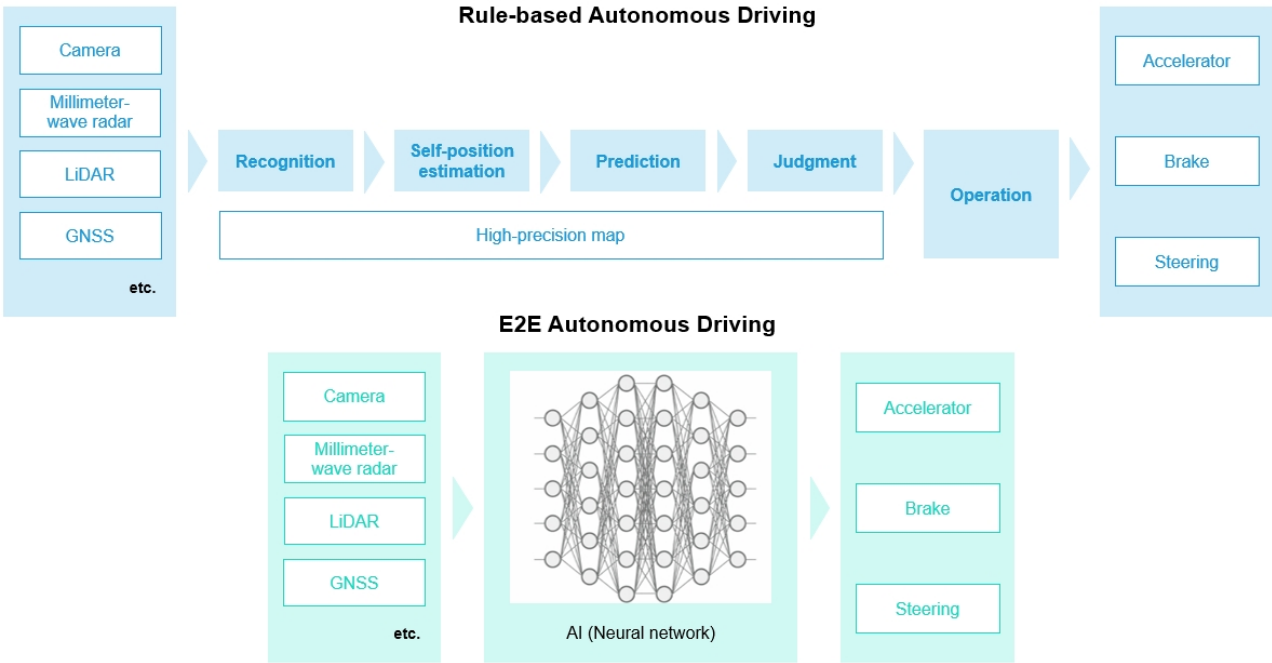
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Rule-based vs. E2E

Autonomous driving systems operate by digitizing perception, judgment, and control through two primary paradigms. The traditional rule-based approach, utilized by players such as Waymo, relies on predefined 'if-then' logic. While this provides transparency for regulatory compliance, it necessitates a costly sensor suite including LiDAR and a heavy dependence on high-definition 3D maps. Consequently, the operational design domain (ODD) of rule-based systems is structurally limited to specific, pre-mapped geographic areas, creating a high barrier to rapid scaling.

In contrast, End-to-End (E2E) systems employ a unified AI model to replicate human-like adaptability (Exhibit 3). Led by Tesla and Wayve, this approach often prioritizes vision-only perception, significantly reducing hardware costs and eliminating the need for constant HD map reliance. By shifting from rigid, human-coded rules to data-driven learning, E2E systems offer superior operational flexibility across diverse environments. However, the primary challenge remains ensuring reliability in rare edge cases that the model has not previously encountered during training.

EXHIBIT 3: Two major approaches to AD: rule-based approaches and E2E approaches



Source: Nikkei, Bernstein analysis

Recent Trends: From Rule-based to E2E, and Toward VLA

Intelligent driving shifting from rule based to end to end

The autonomous driving industry is undergoing a paradigm shift from rigid rule-based logic to End-to-End (E2E) AI architectures, with the next evolutionary step moving toward Vision-Language-Action (VLA) models. Tesla has led this transition by aggressively stripping hardware, removing radar and ultrasonic sensors (USS) to consolidate into a camera-only 'Tesla Vision' system. The 2023 rollout of FSD Beta Version 12 was the definitive turning point, where Tesla replaced millions of lines of hand-coded software with a monolithic E2E model that relies on learned behavior, yielding a step-change in driving performance.

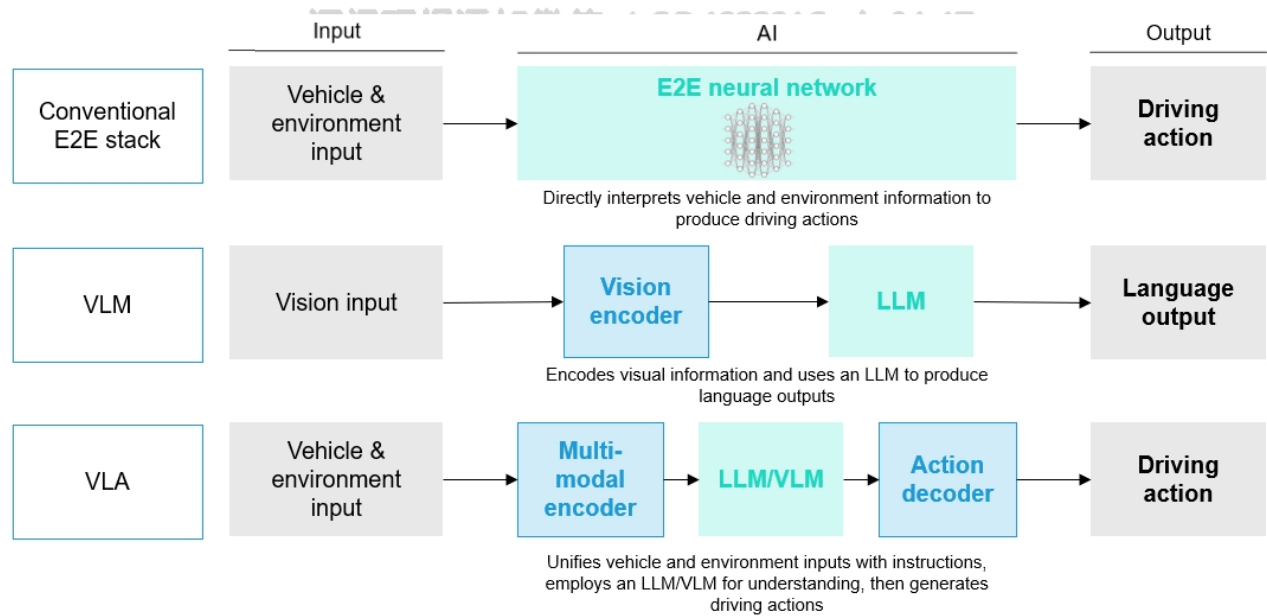
Shift toward VLA within E2E architectures likely to accelerate

Standard autonomous systems often struggle with the 'long-tail' of rare edge cases, such as a human traffic controller overriding a traffic signal at a sudden construction site. To address this, the industry is incorporating Vision Language Models (VLMs), which merge visual perception with generative AI-driven language understanding. Unlike conventional stacks that rely on rigid object detection, VLMs interpret the semantic relationships and intent within a scene. This allows the vehicle to

"understand" complex social contexts and instructions that fall outside the bounds of explicit rule-based programming.

While VLMs are adept at reasoning, they are typically designed to generate text, not vehicle control commands. To bridge this gap, Vision-Language-Action (VLA) models extend the VLM architecture by translating multimodal perception directly into executable driving actions (Exhibit 4). By training on datasets where visual observations and linguistic reasoning are temporally aligned with steering and braking commands, VLA models enable the system to internalize how contextual understanding should inform real-time motion planning. This transition from mere ‘scene description’ to ‘executable action’ is the key to achieving robust generalization in the unpredictable environments required for true L4 autonomy. Among Japanese automakers, only Sony Honda Mobility is actively pursuing in-house VLA development.

EXHIBIT 4: **The E2E domain is shifting toward LLM-powered VLMs and action-generating VLAs**



Source: Turing, Bernstein analysis

Industry accelerating gradually towards L4 & L5 autonomy

As highlighted in [our previous note](#), we forecast that by 2035, L4 and L5 autonomy, which enables a ‘minds-off’ experience where occupants are no longer required to monitor the environment, will reach ~4% of the market. This shift marks a fundamental transition in the industry’s competitive landscape: differentiation is moving away from traditional ‘run, stop, and turn’ driving performance toward the delivery of high-value services through diverse applications. As the driver is liberated from the task of operating the vehicle, the car evolves from a mobility tool into a platform for digital life and monetization.

EXHIBIT 5: **L2+ systems is likely to increasingly being adopted globally across a broader range of vehicle segments**

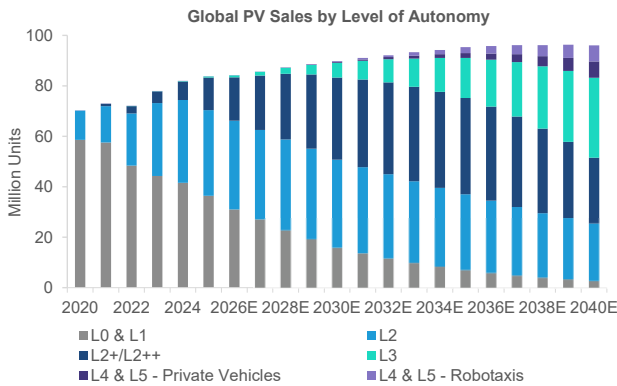
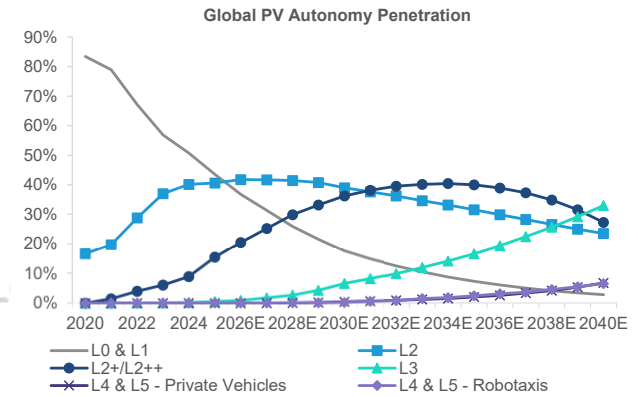


EXHIBIT 6: **Global L2+/L2++ penetration expected to reach 36% in 2030, from 15% in 2025**



3 STRATEGIC DIRECTIONS FOR SDV VALUE CREATION POST-AUTONOMOUS DRIVING

As 'minds-off' L4 & L5 autonomy matures, the automotive industry's competitive landscape will undergo a paradigm shift. Once the driver is liberated from operating the vehicle, new forms of automotive value are created through in-cabin applications rather than traditional hardware performance such as 'run, stop, and turn'. We identify three distinct evolutionary paths emerging among the leaders in the SDV space:

- 1. The Car as a Portable Home (Sharp & Xiaomi):** This direction positions the vehicle as an additional room within the home ecosystem. Sharp's LDK+ concept focuses on stationary utility as a private lounge or office, while Xiaomi's (covered by Eunice Lee) 'Human x Car x Home' vision treats the car as an inseparable node of the smart home. Through HyperOS 2, users can seamlessly manage household tasks—such as adjusting thermostats or even feeding fish—directly from the vehicle's interface.
- 2. Mobility as a Creative Entertainment Space (Sony Honda Mobility):** SHM is leveraging Sony's vast IP ecosystem, including PlayStation, movies, and music, to transform the vehicle into a mobile stage. By utilizing 360 Spatial Sound and high-definition visuals powered by the Unreal Engine, SHM aims to capture the 'attention economy' of transit time, a period previously unmonetized by digital services.
- 3. Robotaxi Monetization (Tesla):** Tesla is shifting the value proposition of the car from a depreciating asset to a revenue-generating investment. Under the vision of 'Sustainable Abundance' (Master Plan Part IV), personal vehicles can join an autonomous ride-hailing network during their typical 95% idle time, fundamentally altering the economics of vehicle ownership.

THE CAR AS A PORTABLE HOME (SHARP & XIAOMI)

Private vehicles typically sit idle 95 percent of the time. With 'minds-off' autonomy, cars evolve into portable living spaces or offices equipped with massive energy buffers. By integrating these with IoT and smart home systems, manufacturers can create a brand lock-in similar to Apple's (covered by Mark Newman) ecosystem as we can see in Phone and AirPods.

Sharp: LDK+ functioning as a new portable room

Sharp (not covered), a Japanese consumer electronics manufacturer, has introduced its LDK+ electric vehicle concept through a strategic partnership with its parent company Foxconn (not covered, owning 57.3% voting rights of Sharp), the world's largest EMS provider (Exhibit 7). The name refers to the 'Living, Dining, Kitchen' layout of Japanese homes, positioning the car as 'Part of your home' rather than just parked there. Evolving from an initial 2024 concept model, the latest LDK+, showcased at the 2025 Japan Mobility Show, utilizes Foxconn's Model A open EV platform and adopts a compact minivan form closer to mass production. Sharp aims to commercialize this vehicle by FY3/28 at a price point comparable to existing family cars. The staff at the Japan Mobility Show 2025 told us that the company was exploring various sales channel options, including partnerships

with major electronics retail chains.

The strategy focuses on maximizing the utility of the vehicle while stationary. Features include 180-degree rotating front seats to create a lounge, a retractable screen, and liquid crystal shutter glass for instant privacy (Exhibit 8). On the software side, Sharp utilizes its CE-LLM (Communication Edge-Large Language Model) platform. This proprietary edge AI technology allows for natural, real-time conversational interaction by processing queries locally on the device rather than relying solely on the cloud. This ensures high responsiveness and data privacy. Combined with AIoT, the car syncs with household appliances like Sharp's refrigerators and washing machines. By integrating with roof-mounted solar panels and V2H (Vehicle to Home) capabilities, the car acts as an intelligent energy management node, further deepening the link between the vehicle and the user's living space.

EXHIBIT 7: Sharp developed the LDK+ concept model based on Foxconn's Model A



Source: Sharp, Bernstein analysis

EXHIBIT 8: LDK+'s features include 180-degree rotating front seats to create a lounge, a retractable screen



Source: Sharp, Bernstein analysis

Xiaomi: Human x Car x Home ecosystem

Xiaomi is pursuing a similar 'Human x Car x Home' vision, where the BEV is the centerpiece of a unified ecosystem (Exhibit 9). One of Xiaomi's key differentiators is the integration of its in-house HyperOS, which underpins products from smartphones and tablets to smart home appliances and cars. This provides a familiar interface where the car's infotainment panel offers the same functionality as a smartphone. Xiaomi even ensures CarPlay compatibility to maintain appeal for iPhone users.

The use cases are highly specific: while at home, owners can prepare the car by defrosting windows and activating climate control. As a user approaches, the car automatically recognizes the smartphone, pairs instantly, and resumes music or navigation. While driving, the interface offers comprehensive control over the smart home, enabling users to adjust thermostats, communicate with doorbell visitors, or even feed fish through connected aquarium devices remotely (Exhibit 10). The system automates home management with location-aware scenarios, such as activating 'Away Mode' when leaving the neighborhood. This level of connectivity treats the car as just another connected appliance, making vehicle choice a byproduct of the user's broader electronics ecosystem.

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EXHIBIT 9: **Xiaomi's 'Human x Car x Home' vision, a smart ecosystem that connects phones, cars and home appliance**



Source: Xiaomi, Bernstein analysis

EXHIBIT 10: **With Xiaomi's HyperOS, home appliances and smartphones are seamlessly connected, enabling users to view home video feeds directly from inside the vehicle.**



Source: Xiaomi, Bernstein analysis

MOBILITY AS A CREATIVE ENTERTAINMENT SPACE (SONY HONDA MOBILITY)

Sony Honda Mobility (SHM), a joint venture between Honda and Sony (covered by David Dai and Robin Zhu), had been preparing to launch entertainment-centric SDVs under the Afeela brand since 2022 (Exhibit 11). However, in March 2026 the company announced that it would discontinue the development and planned launch of its first model, the Afeela 1, as well as its second SUV model. While the highly anticipated products will not come to market, we believe the Afeela concept itself was innovative and clearly illustrated one possible direction for the broader auto industry, namely Mobility as a Creative Entertainment Space. Sony Honda Mobility as a company still remains, and we hope that concepts like this will be revisited in the future.

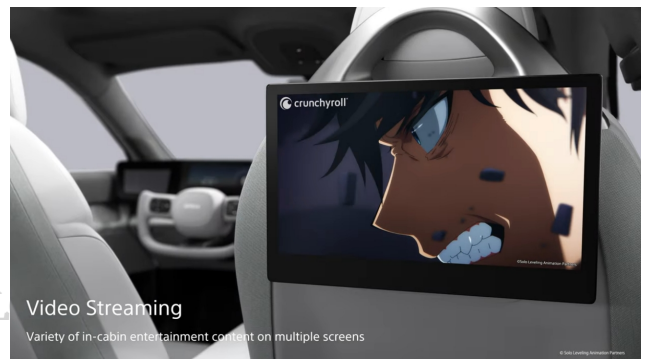
SHM positioned its inaugural model, the Afeela 1, not as a conventional transportation device but as a stage for creative entertainment. Sony sought to leverage its evolution from an electronics manufacturer into a diversified entertainment company to redefine the SDV experience. Specific examples included the integration of movies, PlayStation gaming via PS Remote Play, and karaoke (Exhibit 11). For instance, users would have been able to stream anime via Crunchyroll (Exhibit 12) and play PlayStation titles such as Gran Turismo (Exhibit 13).

EXHIBIT 11: **Sony Honda Mobility has been developing the Afeela 1, set for launch in 2026**



Source: Sony Honda Mobility, Bernstein analysis

EXHIBIT 12: **Users can stream anime via Crunchyroll on Afeela 1**



Source: Sony Honda Mobility, Bernstein analysis

EXHIBIT 13: **Users can also play PlayStation titles such as Gran Turismo on Afeela 1**

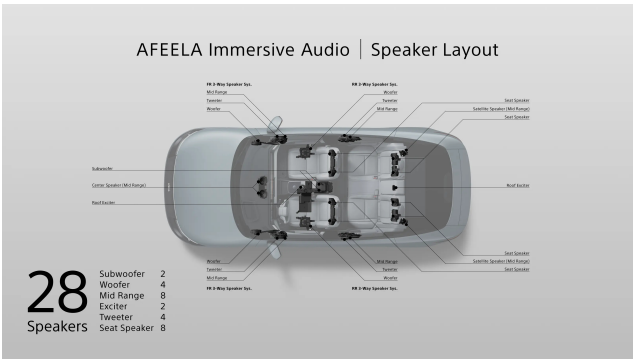


Source: Sony Honda Mobility, Bernstein analysis

To enhance this experience, SHM utilizes 360 Spatial Sound and active noise cancellation technologies, enabled by newly developed 28 speakers leveraging Sony's acoustic technology (Exhibit 14). Sony's expertise handles middle and high-frequency noise, while Honda's technology suppresses low-frequency road noise. This collaborative approach to acoustics creates a studio-quality silence, which effectively elevates the in-cabin entertainment to a premium experience and ensures that high-quality content is delivered as intended.

For ADAS visuals, SHM has moved away from standard tools like Google Maps, instead partnering with HERE Technologies (not covered), the global leader in HD mapping for autonomous driving, and Epic Games (privately held), the video game powerhouse behind Fortnite and the Unreal Engine (Exhibit 15). By leveraging these partnerships, SHM can render surroundings and navigation with a level of visual realism previously reserved for high-end gaming, effectively turning the dashboard into an immersive interface that allows for brand differentiation.

EXHIBIT 14: **The Afeela 1 will be equipped with 28 newly developed speakers leveraging Sony's technology**



Source: Sony Honda Mobility, Bernstein analysis

EXHIBIT 15: **SHM utilizes Epic Games' Unreal Engine for ADAS visualization**



Source: Sony Honda Mobility, Bernstein analysis

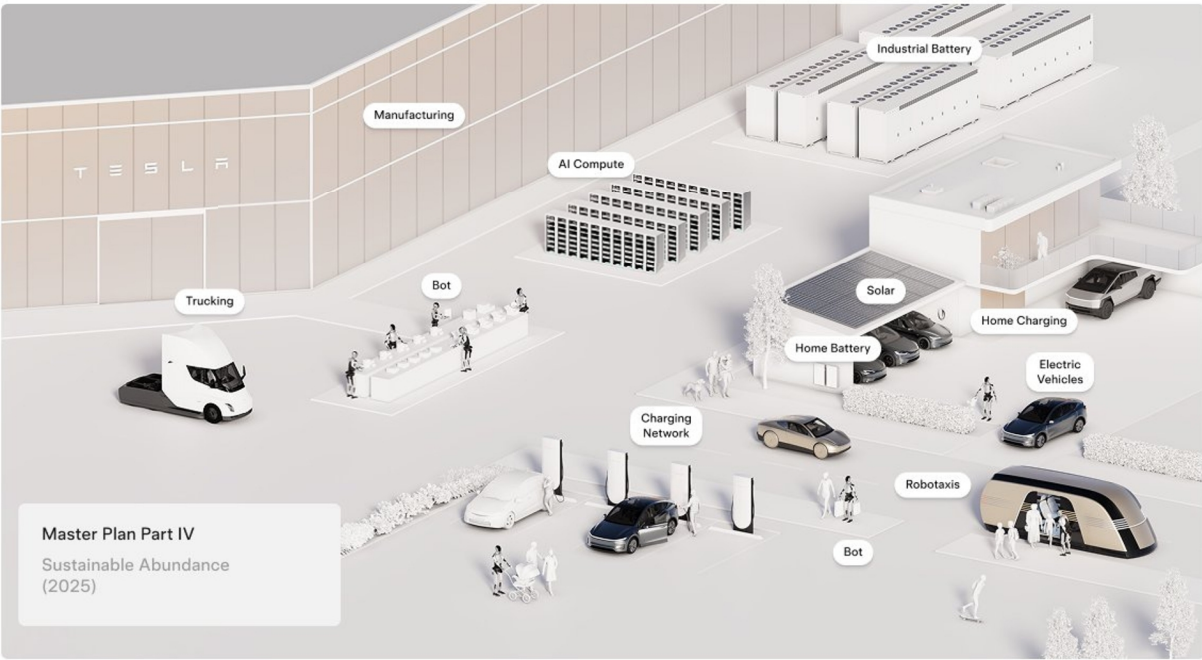
ROBOTAXI MONETIZATION (TESLA)

In a world where full autonomous driving is realized, the 95% idle time of a private vehicle could potentially be monetized as the car works on behalf of its owner. Elon Musk's vision involves a shared economy where personal vehicles join an autonomous ride-hailing network, a concept reinforced in the September 2025 Master Plan Part IV (Exhibit 16). The goal is to flip the script on automotive economics: turning a depreciating asset into a revenue-generating investment for the owner. Tesla has already begun robotaxi operations in Austin and San Francisco using company-owned vehicles and has stated its intention to expand to seven additional US cities in 2026.

Tesla plans to start production of the Cybercab, a dedicated robotaxi with no steering wheel or pedals, at its Texas plant in April 2026 (Exhibit 17, Exhibit 18). If Tesla successfully enables personal vehicles to join this network by next year, it would

fundamentally shift the consumer's value proposition.

EXHIBIT 16: **Tesla's latest vision 'Master Plan Part IV' champions 'Sustainable Abundance'**



Source: Tesla, Bernstein analysis

EXHIBIT 17: **Tesla plans to start production of the steering wheel and pedal-less 'Cybercab' in April 2026**



Source: Tesla, Bernstein analysis

EXHIBIT 18: **The Cybercab is a two-seater featuring a minimalist cockpit with only a central monitor**



Source: Tesla, Bernstein analysis

INTERIOR SPACE AS THE BACKBONE FOR POST-AUTONOMY CABIN EXPERIENCES

With the penetration of Level 4 and Level 5 autonomous driving, the primary differentiator for vehicles is shifting from driving performance to the in-cabin experience. Realizing new mobility concepts such as 'Mobility as a Portable Home' or 'Mobility as a Creative Entertainment Space' requires the advancement of the interior cabin. Specifically, physical flexibility in spatial layout, superior cabin silence, and advanced vibration control serve as technical prerequisites. Unlike competitors that continue to design based on an extension of the traditional driver seat, Toyota and its affiliated suppliers are leading by treating the cabin as an integrated living and activity space.

Toyota's preemptive presentation of interior space concepts

Toyota has consistently introduced original concepts that maximize spatial value ahead of its peers. The e-Palette, a BEV launched in September 2025, is a pioneering example of a modular mobility space where interior units can be swapped based on use cases such as offices, retail stores, or hotels (Exhibit 19, Exhibit 20). For example, when we visited an area where Toyota is operating mobility services using e-Palette, we observed cases in which e-Palette was used not only as a bus but also as a merchandise shop (Exhibit 21). Toyota is currently conducting autonomous driving trials in collaboration with dealers, local governments, and technology partners in specific regions, aiming for a market launch of Level 4 compliant systems by FY3/28. Furthermore, the 6-wheel Lexus LS Concept shown at the 2025 Japan Mobility Show targets the maximization of private space through a redesigned wheel layout that surpasses traditional vehicle packaging constraints (Exhibit 22). The interior utilizes bamboo materials to create a Japanese aesthetic, and the second and third rows can be reconfigured into a face to face arrangement suitable for meetings (Exhibit 23).

EXHIBIT 19: Toyota rolled out the e-Palette in Sep-2025 as a pioneering example of a modular mobility space



Source: Toyota, Bernstein analysis

EXHIBIT 20: The interior space of e-Palette is wide and allows flexible layout configurations



Source: Bernstein photo

EXHIBIT 21: The e-Palette can be used not only as a bus but also as a merchandise shop



Source: Bernstein photo

EXHIBIT 22: The 6-wheel Lexus LS Concept shown at the JMS 2025 targets the maximization of private space



Source: Lexus, Bernstein analysis

**EXHIBIT 23: The second row and third row seats of LS
Concept can be configured to face each other**



Source: Lexus, Bernstein photo

In alignment with Toyota's direction, major interior and exterior suppliers within the Toyota group have clearly defined a transition from individual component orders to integrated system orders in their mid-term business plans.

Toyota Boshoku (not covered) aims to evolve into an 'Interior Space Creator', expanding its capabilities from supplying standalone seats and interiors to planning and proposing entire cabin environments (Exhibit 24). The company plans to increase value add and revenue by securing orders for complete interior systems that integrate cabin controls and acoustic materials such as floor carpets (Exhibit 25).

Toyoda Gosei (not covered) is targeting a 40% revenue share from BEV related products by FY3/31 and is developing specialized functions for autonomous living spaces, including retractable steering wheels and multi-functional consoles (Exhibit 26). By leveraging polymer technology, Toyoda Gosei integrates safety systems and illumination into interior parts to simultaneously achieve design innovation and functional enhancement.

These strategies are realized in concepts like the MX221 for Level 4 ride sharing, showcased at CES 2023, proving the group's ability to provide whole cabin packages by integrating HMI and interior technologies from various affiliates (Exhibit 27). While this advancement serves as a strong driver for increasing ASP, we believe that the group needs to go beyond current collaboration and consolidate development power through capital and/or business restructuring.

As highlighted in [our previous note](#), Toyota Boshoku, Toyoda Gosei, and Tokai Rika each supply different body components for Toyota. As the need for integrated cockpit development grows, we see room for the three companies to pursue collaboration or restructuring/consolidation in the future. Toyoda Gosei is promoting integrated development of safety components such as airbags and seat belts by fully acquiring Ashimori Industry (private), the main seat belt supplier for Mazda and Suzuki, in 2025. In the same year, Toyota reduced its stake in Toyoda Gosei from ~44% to ~20% through a secondary offering, and we believe that a certain level of capital restructuring at Toyoda Gosei has now been completed. On the other hand, Toyota Boshoku, which generates more than 90% of its revenue from Toyota and in which Toyota holds a 32% stake, and Tokai Rika, which derives more than 70% of its revenue from Toyota and in which Toyota holds a 34% stake, do not appear to have completed their restructuring. We will continue to monitor the potential for capital and business restructuring among these three companies.

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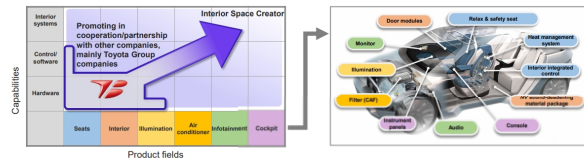
EXHIBIT 24: Toyota Boshoku aims to evolve into an 'Interior Space Creator'

Growing into a Supplier Capable of Planning and Proposing an Entire Interior Space (Interior Space Creator)

By planning and proposing an entire interior space as the Interior Space Creator, increase added value and thereby expand total sales

Expanding capabilities

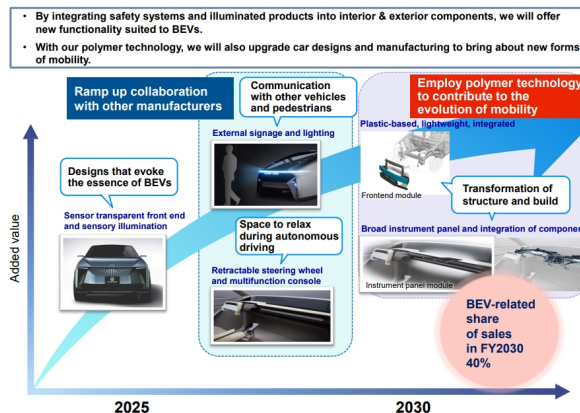
- Acquiring capabilities to receive orders for interior systems (increasing added value)



Source: Toyota Boshoku, Bernstein photo

EXHIBIT 26: Toyota Gosei is developing specialized functions for autonomous living spaces

4. Enhancing Social and Economic Value 4-3. Contribution to Comfort

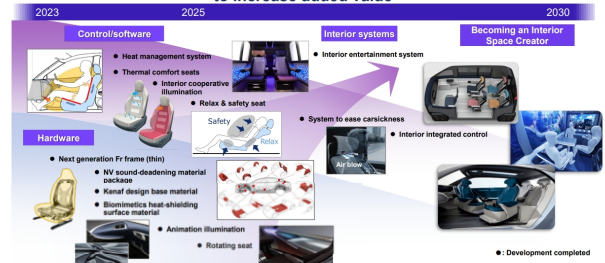


Source: Toyota Gosei, Bernstein analysis

EXHIBIT 25: Toyota Boshoku plans to increase value add and revenue by securing orders for interior systems

Expanding Product Fields (Roadmap)

Connect/integrate various roadmap items and expand each field to increase added value



Source: Toyota Boshoku, Bernstein analysis

EXHIBIT 27: Collaboration among the Toyota interior/exterior suppliers are realized in concepts like the MX221, showcased at CES 2023



Source: Toyota Boshoku, Bernstein analysis

Addressing the shortage of applications at Toyota

While Toyota and its suppliers are making progress in advancing the physical interior space, the company still lacks the application layer necessary to monetize that environment. Following the cancellation of Sony Honda Mobility's Afeela project, the industry lacks a primary driver for in-car entertainment. It is now critical for Toyota to accelerate external partnerships. We believe that expanding specific use cases and service examples to monetize the post-autonomy era will likely be higher priority for the company moving forward.

STRUCTURAL TRANSFORMATION TO BE ACCELERATED TOWARD AI-ENABLED FULL AUTONOMY

As the automotive value chain undergoes a structural transformation toward SDVs and full autonomy, applications are emerging as the primary measure for capturing downstream value. For automakers, the transition to SDVs is not just about connectivity, but about establishing a platform for recurring, high-margin revenue streams that extend far beyond the initial vehicle sale.

Autonomy penetration will accelerate the shift of added value downstream in the car value chain

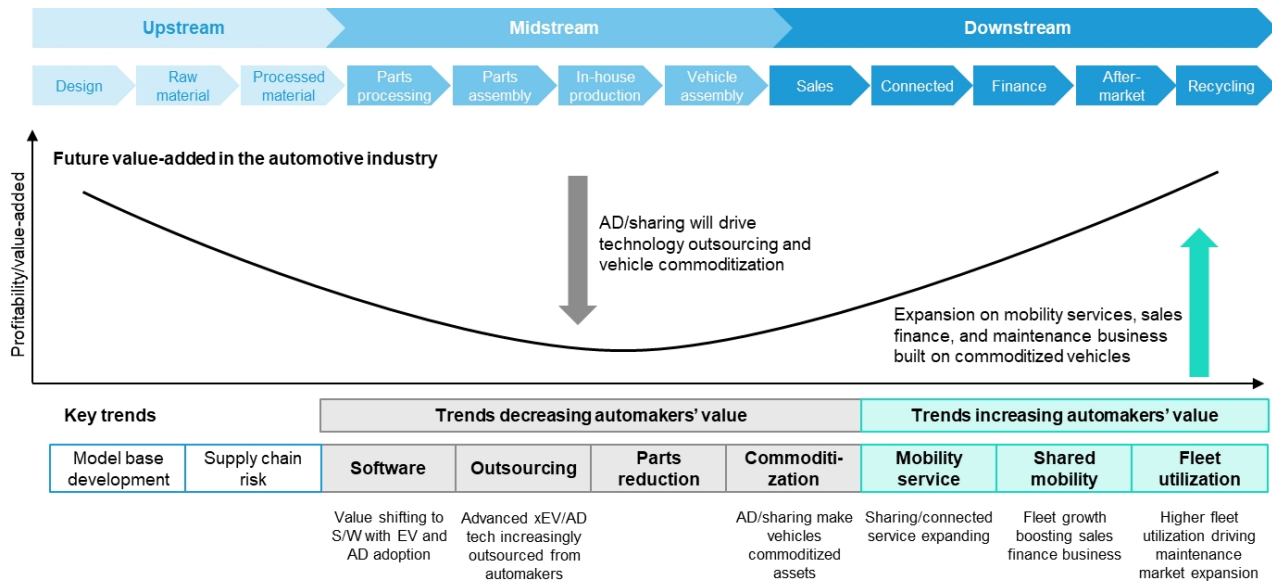
The distribution of added value within the automotive value chain is already beginning to shift. While much of the added value has historically been captured at the center of the chain, where car manufacturers have generated earnings through vehicle production and new-car sales, the value structure is now gradually shifting toward other parts of the chain. As the industry transitions toward electrification, SDVs and more importantly, full autonomy, added value is increasingly moving toward both

upstream and downstream segments, and we expect this trend to accelerate over the coming decade (Exhibit 28).

Upstream, the importance of semiconductors, batteries and rare-earth-based materials is rising as electrification, SDV development and autonomous driving progress. At the same time, technologies embedded in the vehicle are becoming more complex, prompting automakers to outsource an increasing share of high-value domains to external specialists. As a result, the vehicle as a standalone product is becoming more commoditized, mirroring what occurred in smartphones and consumer electronics, and the strategic relevance of traditional manufacturing strengths is gradually diminishing. As AI-enabled autonomous driving accelerates this structural transformation, the move toward E2E and VLA models will further concentrate value in these specialized technology layers.

Downstream, value is shifting toward services built on an expanding installed base. Applications such as connected services, fleet-based mobility solutions, sales finance and high-turnover maintenance operations are becoming more central to monetization models. Indeed, applications could become an important new source of revenue for some traditional automakers, with potential monetization models including direct app sales and subscriptions, transaction fees from external software providers, and even licensing revenue from making applications available to other automakers. Taken together, added value is migrating away from midstream vehicle manufacturing and toward upstream technology domains and, more notably, downstream recurring-service businesses. These trends are already influencing the strategic direction of automakers and will become more pronounced as SDV penetration rises.

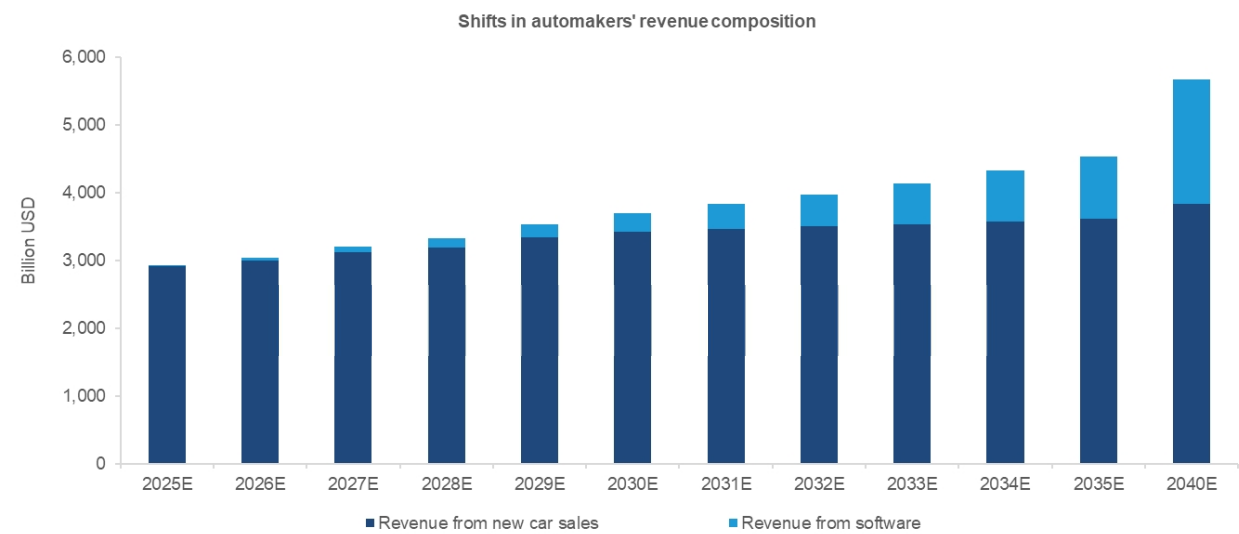
EXHIBIT 28: **Autonomy penetration will accelerate the shift of added value downstream in the car value chain**



Source: The Ministry of Economy, Trade and Industry of Japan, Bernstein analysis

Against this background, we estimate how automakers' revenue mix could evolve over the long term. We model global new-vehicle sales through 2040, the expected penetration of SDVs and smart driving function equipped vehicles, typical vehicle pricing, and subscription fees for connected and ADAS services. Based on these inputs, we derive revenue from new-vehicle sales and from software services. For software, we assume recurring subscription revenue generated from the installed base of SDV and smart-driving vehicles. Today, almost all revenue is derived from new-vehicle sales. However, as SDV and smart-driving penetration accelerates and the installed base grows, we expect software-related revenue to account for roughly one-third of total automakers' revenue in 2040 (Exhibit 29).

EXHIBIT 29: **Automakers' revenue sources to shift from hardware to software**



Source: JEITA, IHS, Bernstein analysis and estimates

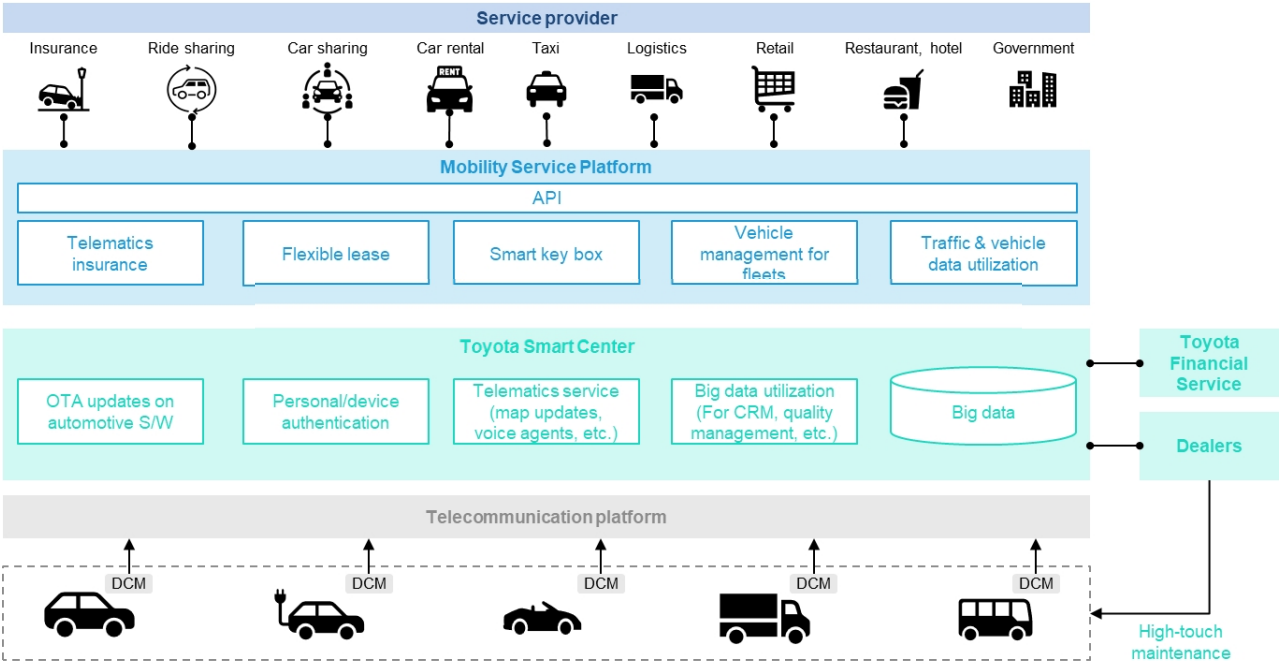
Finally, we expect only a limited number of automakers are likely to be positioned to deliver substantial added value once full autonomous driving is realized. Those with large global vehicle installed bases, scale advantages and sufficient investment capacity, such as Toyota, are the most likely candidates.

Toyota is developing its Mobility Services Platform (MSPF) to enable the global use of driving data (Exhibit 30). MSPF is an open platform offering a suite of functions for mobility-service operators such as ride-hailing, car sharing, rental cars and taxis. Data collected from Toyota vehicles worldwide is aggregated and analyzed through a cloud called Toyota Smart Center and then made available through MSPF. Using this platform, mobility-service providers can develop new service models, and Toyota itself uses the same capabilities for its own new service initiatives. We also believe that MSPF could play a significant role in the future by supporting the collection and accumulation of training data for autonomous-driving applications. Given Toyota's position as the world's largest automaker by new vehicle sales and as a manufacturer with one of the largest global vehicle fleets at about 150 million units, the company holds a substantial advantage in both the volume and the diversity of driving data it can capture.

Mid-sized and smaller manufacturers may find it increasingly difficult to secure downstream value and maintain long-term competitiveness, as weakening value capture leaves them at risk of remaining just vehicle manufacturers. Nissan, Wayve and Uber (covered by Nikhil Devnani) announced Japan's first autonomous driving project, under which Wayve will provide its HD map-free E2E stack 'AI Driver' and Nissan will supply the vehicles (Exhibit 32).

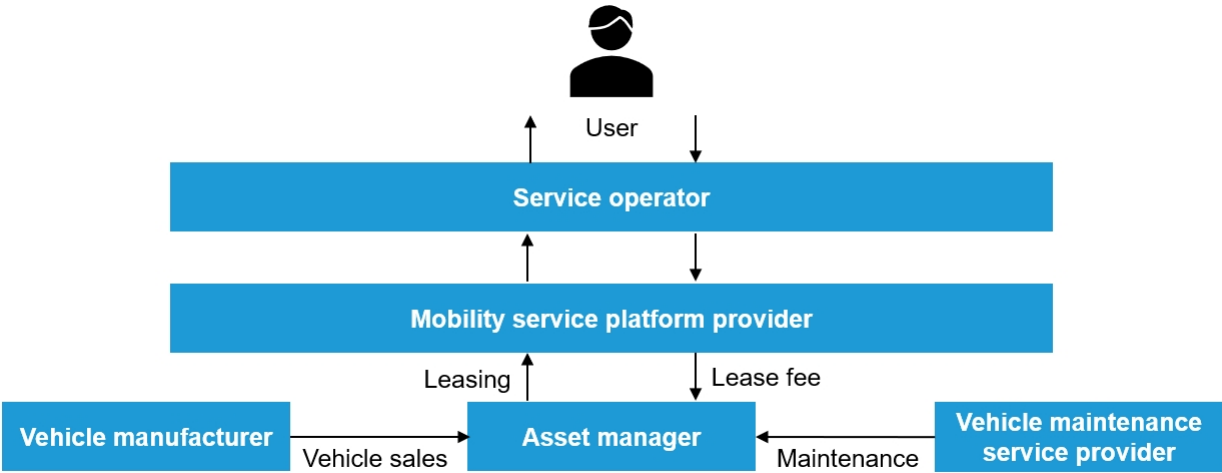
While such headlines appear promising at first glance, a more critical examination is required regarding automakers' long-term differentiation. In this ecosystem, Wayve controls the core autonomous driving intelligence, while Uber owns the direct customer interface and service operation. This might leave Nissan primarily as a hardware supplier, raising concerns about how Nissan will differentiate their vehicles from competitors when the important software and service-related value is captured by their partners. Without a clear strategy to reclaim a share of the downstream value, traditional manufacturers risk being marginalized into low-margin assembly roles within the new mobility landscape.

EXHIBIT 30: **Toyota developing its ‘Mobility Service Platform (MSPF)’**



Source: Toyota, Bernstein analysis

EXHIBIT 31: **Business domains where automakers can play will likely vary**



Source: Bernstein analysis

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EXHIBIT 32: **New Nissan Leaf produced for pilot robotaxi projects in Tokyo by Uber**



Source: Nissan, Bernstein analysis

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